

Experiments with voice modelling in speech synthesis

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Abstract. Some experiments with voice modelling using recent developments of the KTH speech synthesis system will be presented. A new synthesizer, GLOVE, an extended version of OVE III has been implemented in the system. It contains an improved glottal source built on the LF voice source model, some extra control parameters for the voiced and noise sources and an extra pole/zero-pair in the nasal branch. Furthermore, the present research versions of the KTH text-to-speech system include possibilities for interactive manipulations at the parameter level with on-screen reference to natural speech. The synthesis system constitutes a flexible environment for voice modelling experiments. The new synthesis tools and models were used for synthesis-by-analysis experiments. A sentence uttered by a female speaker was analysed and a stylized copy was made using both the old and the new synthesis system. With the new system the synthetic copy sounded very similar to the natural utterance.

Zusammenfassung. Es werden einige Experimente vorgestellt welche die Modellierung des Glottissignals betreffen und welche neue Entwicklungen des KTH Sprachsynthesystems ausnützen. Ein neues Syntheseverfahren, GLOVE, welches eine erweiterte Version des OVE III ist, ist in das System eingebaut worden. Es enthält eine verbesserte Darstellung des Glottissignals welches auf dem LF Quellenmodell beruht, einige zusätzliche Kontrollparameter für die stimmhaften und stimmlosen Quellen und eine zusätzliche Pol/Null Stelle in dem nasalen Seitenkanal. Desweiteren enthält die heutige Entwicklungsversion des KTH Text-zu-Sprache Systems Möglichkeiten um die Parameter interaktiv zu verändern unter Bezugnahme auf natürliche Sprache. Das Synthesesystem stellt eine flexible Umgebung dar um Experimente mit Stimmtonmodellen durchzuführen. Die neuen Syntheseverfahren wurden benutzt um Synthese-durch-Analyse Experimente auszuführen. Ein von einem weiblichen Sprecher gesprochener Satz wurde analysiert und es wurde eine stylisierte Kopie hergestellt mit dem alten und dem neuen Synthesesystem. Mit dem neuen System klangen die synthetische Kopie und die natürliche Produktion sehr ähnlich.

Résumé. Dans cet article, des expériences de modélisation de voix utilisant une version récente du système de synthèse de parole du KTH sont présentées. Un nouveau synthétiseur, GLOVE, version étendue de OVE III, a été implémenté dans le système. Il contient une source glottique améliorée basée sur le modèle LF de la source vocale, quelques paramètres supplémentaires de contrôle des sources voisée et bruitée, ainsi qu'une paire pole/zero pour le branchement nasal. De plus, les versions du système de synthèse texte-parole présentées dans cet article rendent possibles les manipulations interactives au niveau des paramètres avec référence sur écran à la parole naturelle. Le système de synthèse constitue donc un environnement flexible pour les expériences de modélisation de voix. Les nouveaux modèles et outils de synthèse ont été utilisés dans des expériences de synthèse par analyse. Une locutrice produisait une phrase dont on effectuait une copie stylisée qui employait respectivement l'ancien et le nouveau système de synthèse. A l'écoute, la copie synthétique effectuée avec le nouveau système ressemble fort à la parole naturelle.

Keywords. Speech synthesis, voice modelling, voice source models, text-to-speech conversion, female voice.

1. Introduction

The need for voice variations is apparent in different speech synthesis applications, such as voice prosthesis and translating telephony. Speaking style variations are an important means of dis-

criminating information of different kinds (Bladon et al., 1987). Data on speaker variability is now being accumulated. Fant et al. (1990) have investigated speaker variations in the context of a multi-talker speech data base. More detailed analysis of voice source dynamics have been per-

formed by Gobl (1988) and Karlsson (1988). The KTH text-to-speech system has recently undergone substantial modifications to meet the needs for variations on the synthesized voice. The present synthesis system contains a synthesizer that can utilize the new knowledge about speaker variations. This extended synthesizer, called GLOVE, contains several new features compared to the standard OVE III (Liljencrants, 1968), including an expanded LF voice source model (Fant et al., 1985). GLOVE is presented in Figure 1. The LF-model is a more realistic voice source than the filtered impulse approximation of the voice source in OVE III. The spectral properties of this new source and the possibility of dynamic variations have proved to be essential for modelling a female voice.

The new speech synthesis system has been used as a research vehicle to study both global effects, voice transformations and more individualized transforms implemented by changes in definitions and rules in the text-to-speech system. The present research versions of the KTH text-to-speech system in combination with the possibility for interactive manipulations at the parameter level with on-screen reference to natural speech constitute a flexible environment for such experiments. Special effort is invested into the creation of a female voice. Transformations by global rules of male parameters are not judged to be sufficient.

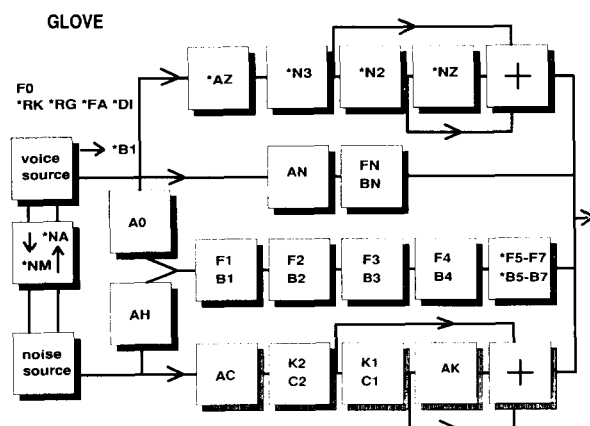


Fig. 1. Configuration of the extended synthesizer GLOVE. The new parameters are marked by *. For explanations of the parameters, see the text.

2. Improved synthesizer

For several years we have worked with a text-to-speech system that uses a version of the now classical OVE III cascade formant synthesis model and a simple filtered impulse glottal source. The virtues of the simple source model have been easy parametric control and straightforward hardware implementation. The drawbacks have been the somewhat ad hoc rules to simulate the essentials of segmental glottal variations by means of bandwidth and amplitude control. We have also experienced limitations when modelling different voices. While the male voice is regarded as generally acceptable we believe that an improved glottal source will to some extent open the way to more realistic synthesis of child and female voices as well as creating more naturalness and variations in male voices (Carlson et al., 1989a; Karlsson, 1989).

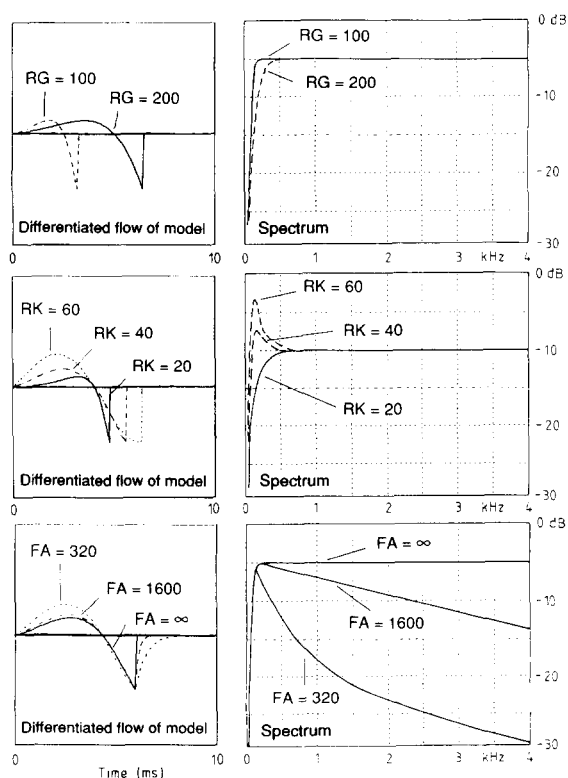


Fig. 2. The influence of the parameters RG, RK and FA on the differentiated glottal flow pulse shape and spectrum. (From (Gobl and Karlsson, 1989).) The spectra are preemphasized by 6 dB/octave.

With emphasis on signal processor realizable solutions, different versions of the LF-model have been tested, including storing a subset of the pulse shapes that this model generates. One version of the voice source has been a truncated exponential sinusoid followed by variable cutoff -6 dB/octave low-pass filter modelling the effect of the return phase, i.e. the time from maximum excitation of the vocal tract to complete closure of the vocal folds. This version has computational attractions but the relations to analysis parameters are to some extent compromised. The present solution is the full two-part waveform generation suggested by Fant et al. (1985). The control parameters used for this source are RK, RG, EE, FA and F0. RK corresponds to the quotient between the time from peak flow to excitation and the time from zero to peak flow (in the differentiated flow in Figure 2 this corresponds to the time from the zero-crossing to the negative peak divided by the time from the start to the zero-crossing). RG is the time of the glottal cycle divided by twice the time from zero to peak flow. RG and RK are expressed in percentages. EE is the excitation strength in dB and FA the frequency above which an extra -6 dB/octave is added to the spectral tilt. RG and RK influence the amplitudes of the two to three lowest harmonics, FA the high frequency content of the spectrum and EE the overall intensity. Figure 2 explains the function of these parameters.

The synthesizer has been supplemented with a flow control of the noise source. The glottal flow waveform is used to amplitude modulate the noise. This source is used to model frication at supraglottal constrictions in the vocal tract. It is connected to the vowel branch and the parallel pole zero branch used for the synthesis of fricatives. One major advantage of the flow modulated noise is the more convincing way to synthesize voiced fricatives. The degree of this modulation can be controlled by the parameter NM.

A new parameter NA has been introduced to control the mixing of noise into the voice source. The noise is added according to the glottal opening and is thus pitch synchronous as is the noise control by the NM parameter. The NA parameter models noise generated at the glottis, while NM describes supra-glottal noise generation. Either

the noise source is flow modulated (NM), or noise is added to the voice source (NA). Similar methods were also used in the source model that we explored in earlier experiments (Rothenberg et al., 1975).

Another vocal source parameter of a slightly different function is the DI parameter with which creak, laryngalization or diplophonia can be simulated. We have adopted a strategy discussed in the paper by Klatt and Klatt (1990), where every second pulse is lowered in amplitude and shifted in time.

The "source-vocal tract" is not implemented in the current synthesizer. The only interaction control concerns the bandwidth of the first formant that is adjusted according to glottal opening. The inclusion of this feature seems to add to the naturalness of the speech in the synthesis experiments performed. It is however not clear whether this parameter has any perceptual advantage compared to a simple setting of an average bandwidth. Experiments by Nord et al. (1986) did not show a significant effect on this issue.

The synthesis of nasals has been a problem in our work to improve the segmental quality. In the new synthesis model, two new poles, N1 and N2, and a zero, NZ, have been added in parallel to the original nasal resonance to solve some of these problems. In a perceptual evaluation, about 2/3 of the nasal feature errors disappeared after the addition of this new branch (Carlson et al., 1990a).

In the simulation of a female voice the treatment of higher poles has to be more flexible compared to earlier models. A higher pole correction consisting of three formants that correspond to the fifth, sixth and seventh resonances has been introduced. The fifth formant is controlled like the lower formants, while the following two have a fixed relation to this formant. Preliminary these are put at a ratio of 1.2 and 1.4 relative to the fifth formant. The spectral peak corresponding to the seventh formant will not be prominent after the low-pass filtering (6.3 kHz) at the output. In the old synthesizer, the higher pole correction consisted of three poles of the same frequency, that could be controlled like the lower formants, and with fixed bandwidths.

One important origin of distortion in a digital

synthesis model is the updating of parameters. In the old synthesizer the filter coefficients were updated pitch synchronously according to the last frame from the control program (10 ms frame update). In the new synthesizer different strategies were tried. Eventually, linear interpolation between consecutive control frames at the time for update was chosen. This interpolation is very important to avoid unwanted distortion for a high pitch voice, that would appear if the parameter configuration was kept constant for two consecutive glottal pulses. To evaluate the importance of an even more frequent update, a sample by sample update of filter coefficients was implemented. The method was abandoned as no significant difference could be found between this method and the previously described method.

The noise source amplitude is updated frame by frame. If the source amplitude is zero, the noise generator is reset. In this way the same type of noise excitation is used at the release of every stop and the place cues can be modelled more reliably.

The new synthesizer has been integrated in the control framework of the text-to-speech system. Some of the generalizations from the analysis work are now being implemented and evaluated.

3. Development tools

The central part of the development facility is the rule development environment, where rules can be formulated, compiled and tried out in separate rule components (Carlson et al., 1990b). Trace facilities of different kinds exist, from simple printouts of the results to detailed traces of individual rule applications. Statistics of rule productivity for example can also be gathered. On the parameter level, graphs of parameter tracks can be displayed. Support programs for maintaining frequency ordered reference word corpora have been developed. The development environment is implemented on several computer systems including IBM/PC compatibles and most recently on our network of Apollo work stations. The Apollo system is also used for general speech analysis, and it has been possible to combine speech analysis and synthesis in an environment

efficient for developing the phonetic component of the text-to-speech system.

One important component for synthesis development is the rule controlled data base search system (Carlson et al., 1989b). The system accesses label files in our speech data base. These label files have the phonetic transcriptions stored together with the time location of all segment boundaries. A search-rule system, written in the RULSYS notation, uses these files as input. Different phonemes in specific positions can thus be marked and stored for examination. We can, for example, formulate a rule that selects all stressed /i:/ vowels. These vowels can then be analyzed and presented in a contour histogram representation. The bottom graph in Figure 3 is an illustration of this method. The synthesis model is compared to the contour histogram of 83 vowels produced by one male speaker. Since the higher pole correction is of the older type described above, it needs to be placed at a rather high frequency. The higher formants will otherwise get too much emphasis.

An important complement to the rule generated synthesis is manually controlled synthesis, so called HI-FI synthesis or "analysis by synthesis". An extension of RULSYS, called HISYS, uses rule generated parameter data as input and allows handediting of these data. This gives the user a quick start and also creates a link between the two approaches. The phonetic transcription of an utterance is entered in RULSYS and the rule-controlled parameter tracks are sent to the HISYS component where they are plotted on the screen. One option in HISYS allows a spectrogram of natural speech to be displayed superimposed on these parameter tracks using the same frequency and time scales. The synthesis parameter values can then be adjusted according to the natural utterance. The experiments described below have been carried out using this option.

With the help of a cursor, each parameter can be adjusted, parameter values can be deleted or new parameter values inserted. During the whole process, the synthesis can be listened to in an interactive way. The synthesized sound can also be compared to earlier versions. The user might study the spectral shape at a specific time frame in the synthesis. The time frame is marked and a

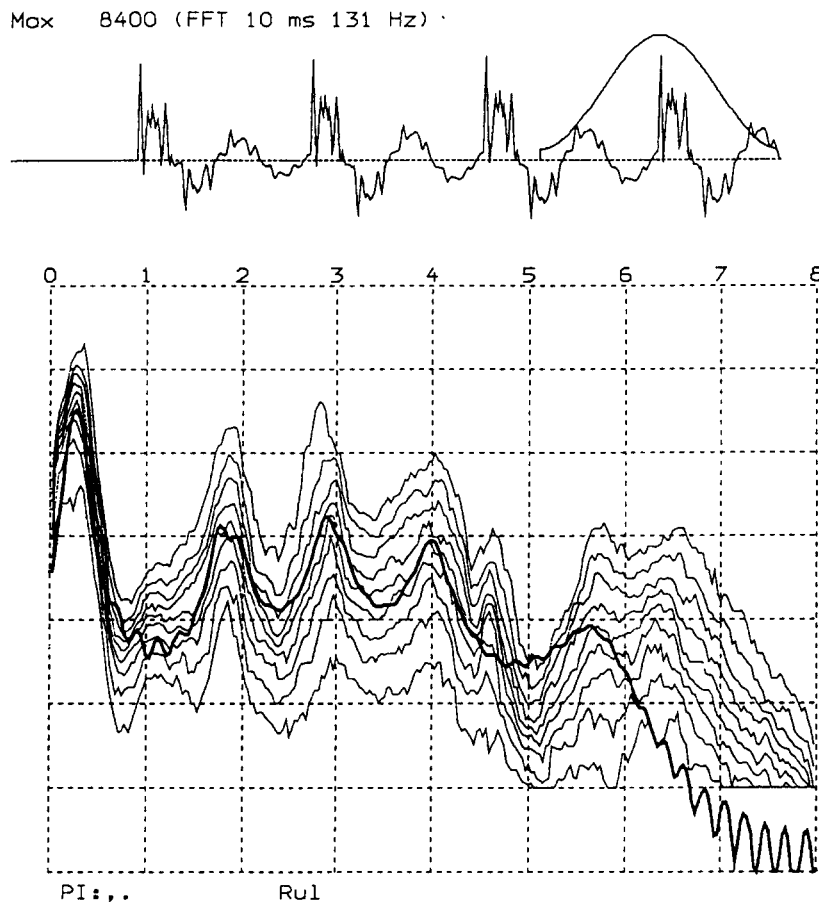


Fig. 3. Upper part: Time wave and analysis window for the synthetic vowel /i:/. Lower part: Contour histogram representation of the energy distribution for 83 /i:/-vowels (thin lines) and the spectrum for the synthetic vowel /i:/ (thick line).

software simulation of the synthesizer is used to create the corresponding spectral section. This section can be compared to natural speech or, as an alternative, the result of a data base search as is illustrated in Figure 3.

4. Experiments with voice quality

In order to evaluate the new synthesizer it was used to create different voices. Our approach was to make a stylization of individual utterances spoken by speakers with different voice characteristics. This process started with rule generated parameters and the target values were adjusted according to the different voices, using the possibility to superimpose a spectrogram of natural

speech on the speech synthesis parameter traces. Results from inverse filtering provided the appropriate voice source parameters. Typically one or two specifications per speech sound has been used for each vocal tract and source parameter, i.e. in contrast to the approach taken by Pinto et al. (1989), we are not trying to model the human speaker frame by frame, but rather to make a stylization that later on can be generalized when formulating rules for different speakers. The stylization approach was employed to produce a synthesis reference with an acceptable female voice based on a recording of a female speaker. A spectrogram of the natural Swedish sentence "Pia odlar blå violer", used in this synthesis experiment, can be seen in Figure 4. It contains a fair amount of voiced segments which is of in-

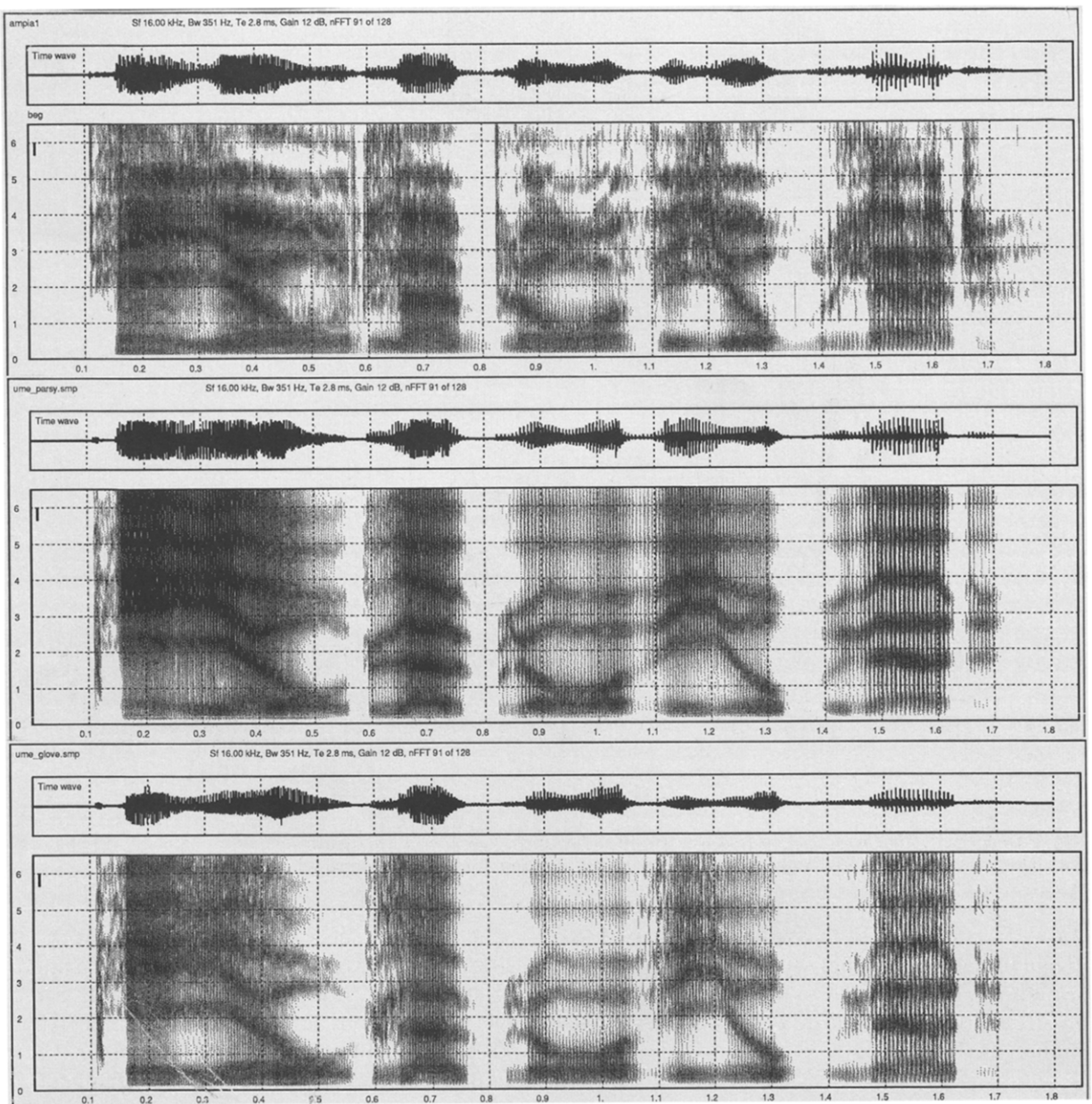


Fig. 4. Time wave and broad band spectograms of the Swedish sentence "Pia odlar blå violer", /pi:a u:dlar blo: vlu:ler/ uttered by a female speaker (top), produced with the old synthesis model (middle) and produced with the new synthesis model (bottom).

terest for a test of the new glottal source.

The sentence was analyzed in terms of segment durations, formant frequencies, fundamental frequency and glottal source parameters. In the experiment, a linear interpolation between turning points for the formant parameters was used. A slightly different method is chosen for the rule

synthesis program; the linear smoothing is described by a target value and a time to reach this value. Both target and time can be specified by rule. This has some advantages since it is less dependent on the specific duration structure. However, in a synthesis by analysis experiment the linear method is more suitable.

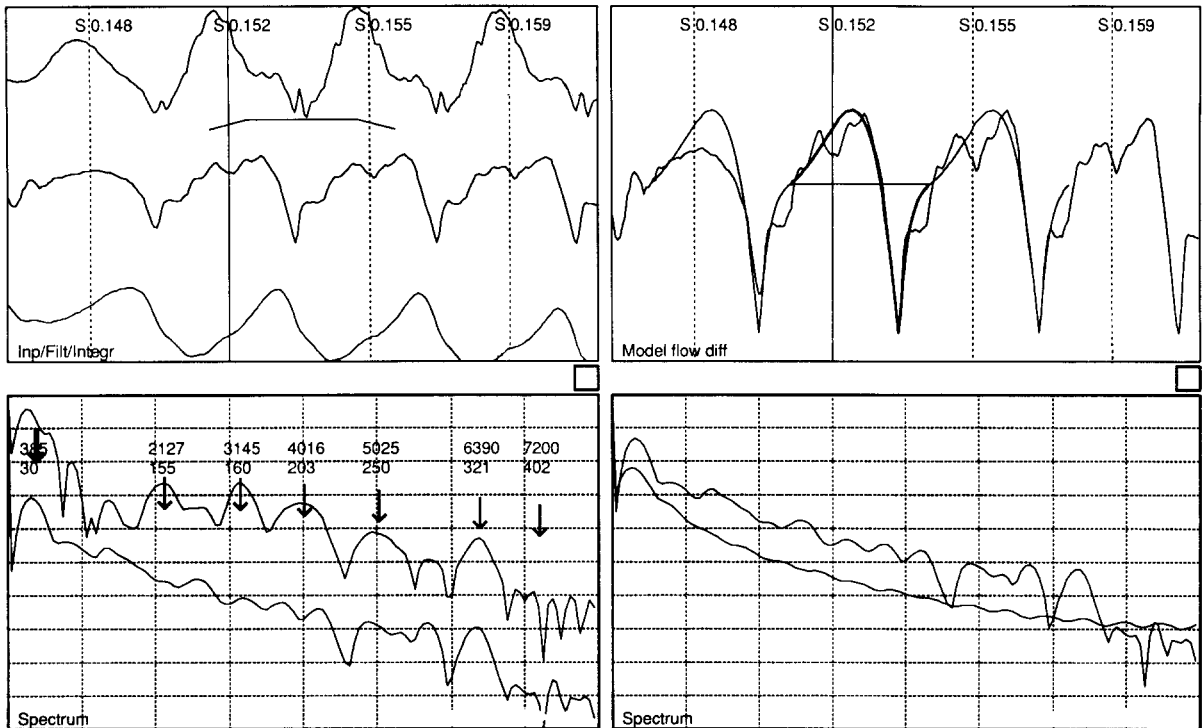


Fig. 5. Example of inverse filter analysis. The left upper part shows from the top: the speech wave, the inverse filtered speech wave and an integrated version of it. Below this the spectrum of the speech and the inverse filtered speech is shown together with the filter settings. To the right a matching of the LF-model to a voice pulse is shown, with the time wave above and the corresponding spectra below.

The measured specifications were used as control parameters for the old synthesis model to create a synthesis stimulus. A spectrogram of this stimulus is displayed in Figure 4. As can be seen, the synthesis deviates considerably from the natural sentence. Most striking is the high degree of aspiration in the voiced segments for the natural speech which is not modelled in the synthesis.

The sentence was inverse filtered at a few positions in each phoneme, typically at the boundary between two phonemes, in the middle of a phoneme and in the most constricted part of the /u:/, /o:/ and /i:/ vowels. A voice pulse defined by the LF-model parameters described above was fitted to the inverse filtered pulse to get the source parameters for the synthetic voice source. An example of the analysis of the /i:/ vowel is shown in Figure 5. (The same vowel /i:/, uttered by a different speaker is shown in our database example, Figure 3.)

The voice source specifications were added to the specifications used for the old synthesis model and an utterance was produced using the extended synthesizer GLOVE described above. It was found that though the quality of the synthesis was much improved, the improvement was not sufficient to achieve a good female voice quality.

Several aspects had to be modelled in addition to the formant values and the voiced source parameters. The new higher pole correction proved to be a valuable improvement for the spectral balance, but the default bandwidth of the first formant was too narrow. (Without adjusting this value the synthesis was perceived as speech produced in a resonant room.) The bandwidth of the first formant was adjusted according to the glottal opening rather than given the same high value for the whole pulse.

The major difference between our speaker and the synthesis was the breathy source. The noise had been a problem already in the inverse filter-

ing phase. The NA parameter was used to control an addition of pitch synchronous noise. The addition of noise was specially important to mark the juncture between the /i:/ and the /a/ and in the final part of the sentence.

Addition of noise in the vocalic consonants in some contexts was also of importance. However, this type of noise is probably of a different nature and can often occur also in voices with very little glottal leakage in other contexts. The frication in an /l/ sound after a stop is common but can be lacking if a morph boundary is intervening.

A spectrogram of the adjusted synthesis is shown in Figure 4. Several aspects have been improved and the quality is significantly better than the first version of the synthesis. The segmental details can be adjusted for some of the consonants, but our major concern is the modelling of the constricted phase of the rounded vowel /u:/. The high frequency energy is still too low and can only be compensated for by adjusting the vocal source in an ad hoc way.

5. Conclusions

Modelling a wider range of speaker styles is increasingly important in speech synthesis. In this paper we have described how the new version of the KTH text-to-speech system can be used for pursuing such studies. In the new version, many of the restrictions in the original OVE III model have been alleviated. The improved speech production model, GLOVE, has proved its value in modelling a female voice. We believe that speech synthesis experiments are an important tool in investigating speaker variability. It remains to be seen if the increased flexibility of the synthesis model will suffice to model a wider range of voices, including pathological voices.

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